Air-Quality Impact of Sustainable-Mobility Scenarios in Guimarães

OpenFOAM Hackathon Challenge - OFW21

CFD assessment and PODI reduced-order modelling of CO and NO_x at four sensitive receptors

Team 3

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Executive summary

Four mobility scenarios: the provided reference case, two fleet-electrification scenarios (S1: 20% of gas vehicles replaced by EVs; S2: 40%), and a Metro-Bus corridor scenario (S3). These scenarios were evaluated with a steady-RANS dispersion workflow on the real terrain of Guimarães. Pollutant concentrations of CO and NO_x were sampled at the four sensitive receptors defined by the challenge: the Public Hospital and the Martins Sarmento, Francisco de Holanda and Santos Simões high schools. To extend the assessment beyond the ten directly simulated hours, the CFD snapshots were then used to train a Proper Orthogonal Decomposition with Interpolation (PODI) reduced-order model, which reconstructs full concentration fields for any wind and emission condition at negligible cost and enables the complete 24-hour exposure cycle to be evaluated.

Both the direct CFD results and the surrogate agree on the central findings: electrification lowers concentrations uniformly at every receptor (-20% for S1, -40% for S2), whereas the Metro Bus produces a strongly location-dependent benefit - comparable to S2 at the Public Hospital, which borders the reduced corridor, but only a few percent at Santos Simões, the most exposed site. The exact -20/-40% scaling of the electrification cases also serves as a consistency check on the modelling chain, and the surrogate reproduces held-out CFD fields with a parity R^2 of 0.96. No single measure removes the Santos Simões hotspot, which points to a combined global-plus-local strategy.

1. Objective

The challenge is to quantify how sustainable-mobility measures change urban air quality at four sensitive locations, using the provided reference case as the baseline and reporting the expected air-quality improvement of each scenario. The assessment metric is the pollutant concentration sampled at the receptor surfaces (ROI), reported per pollutant and compared against the reference both in absolute terms and as a percentage change. A second objective is to turn the resulting CFD dataset into a practical reduced-order model that supports rapid scenario screening, receptor-level exposure assessment, and evaluation of the full daily cycle without repeating the CFD workflow.

2. CFD methodology

Domain and mesh. The provided terrain and region-of-interest building surfaces are recentred to the origin and meshed once with snappyHexMesh on a COST-732-style background domain; the same mesh, boundary conditions and solver settings are reused for every hour and every scenario, so only the emission input changes between scenarios and comparisons are fair.

Flow. The hourly wind field is computed with steady RANS (simpleFoam, k-ε with neutral atmospheric-boundary-layer wall functions). An all-round atmospheric inlet accepts the hour-varying wind direction; a first-order warm-up followed by a second-order restart keeps the solve stable on the steep terrain. The converged U, φ and turbulent viscosity are frozen and passed to the dispersion stage.

Dispersion. Each pollutant is transported as a separate passive scalar (scalarTransportFoam) on the frozen flow, with the road network carved into the ground as a dedicated 'streets' patch carrying a non-uniform fixed-gradient flux derived from the per-segment hourly emission factors. CO and NO_x are solved independently and reported separately. Concentrations are therefore the local traffic increment, with no regional background added.

Receptors and temporal strategy. Concentration is sampled as the average area over each of the four ROI receptor surfaces (surfaceFieldValue function objects). To keep the 24-hour problem tractable while preserving variability, ten representative hours were selected (maximin over wind speed/direction and emission totals) and run identically for all four scenarios. The whole sweep is orchestrated with a Snakemake workflow that keeps the large mesh decomposed end-to-end on the HPC system, reusing each hour's flow across all scenarios. The full 24-hour cycle is then covered by the reduced-order model described in Section 4.

3. Scenarios

The three mobility scenarios are implemented purely as a scaling of the per-segment emission factors (the flow is identical across scenarios). S3 applies the reductions to the street segments listed in road_ids_reduction.txt. These were verified to map correctly onto the emission rows, and corresponding predominantly to the Circular Urbana ring together with one EN101 section, i.e. the bus corridor rather than the full EN101. Because the corridor segments are reduced by different amounts (between 30% and 50% depending on the segment), the local intervention is not a single uniform scaling. For the surrogate, each scenario is therefore expressed through two compact labels: a global emission factor G applied to all segments, and a categorical corridor indicator $L \in \{0, 1\}$ that flags whether the Metro-Bus reduction on the tagged roads is active.

Scenario	Definition	Emission scaling	G (global)	L (corridor)
Reference	Provided traffic (baseline)	×1.00	1.0	0 (off)
S1	20% of gas vehicles → EV	×0.80, all segments	0.8	0 (off)
S2	40% of gas vehicles → EV	×0.60, all segments	0.6	0 (off)
S3	Metro Bus (Guimarães–Braga corridor)	×0.50 / ×0.70 on listed segments	1.0	1 (on)

Table 1. Mobility scenarios: emission scaling used in the CFD runs and the corresponding surrogate labels. L is a categorical indicator of the corridor intervention, not an exact reduction factor.

4. Reduced-order model (PODI)

Proper Orthogonal Decomposition with Interpolation (PODI) approximates high-dimensional simulation fields with a small number of dominant spatial patterns. The patterns are learned from the existing CFD snapshots, each full concentration field is represented through a compact set of modal coefficients, and those coefficients are interpolated for new operating conditions. Instead of running a new full CFD simulation for every hour, wind condition, or emission scenario, the PODI model reconstructs the concentration field from the learned modes- making full-day pollutant behavior and scenario comparison possible at a small fraction of the computational cost.

4.1 Input data and labelling

Snapshots. The training data consists of the converged pollutant concentration fields obtained for the reference condition and the traffic-reduction scenarios of Section 3. Each snapshot represents the spatial concentration field for one pollutant at one hour; the representative hours cover variations in wind speed, wind direction, and emission strength.

Labels. Each snapshot is associated with a compact physical parameter vector p = (u, v, G, L), where (u, v) are the two horizontal wind components, G is the global emission scaling factor, and L ∈ {0, 1} is a categorical indicator that flags whether the local corridor intervention is active (Table 1). L is deliberately not an exact reduction factor: the tagged road segments are reduced by varying amounts (30–50%), so the net effect of the intervention is left for the model to learn from the snapshots rather than being imposed as a fixed scaling. This parameterization still makes it possible to predict cases with and without the corridor measure.

Training and testing. The available snapshots are split into training and test sets while preserving all scenario types in both sets. The same split is used for each pollutant so that the surrogate accuracy can be compared consistently.

4.2 Method and approximation function

Magnitude–shape decoupling. Concentration magnitude varies strongly (roughly 50× across hours) because pollutant levels depend on both wind dilution and emission strength. To avoid forcing a single linear model to learn field magnitude and spatial pattern at once, each snapshot x_c is split into a scalar volume-weighted magnitude and a unit-norm shape:

$$s_c = \|x_c\|_V = (x_c{}^T V x_c)^{1/2}, \quad \hat{x}_c = x_c / s_c, \quad V = diag(cell\ volumes)$$

Here s_c is "how much" pollutant is present and $\hat{x}_c$ is "where" it sits.

POD by the method of snapshots. POD is applied to the normalised shapes. The method of snapshots forms a small Gram matrix from the available fields and extracts the dominant modes without explicitly storing a full high-dimensional basis, which makes the approach feasible even when the CFD fields contain many millions of cells.

Approximation function. For a new parameter vector p = (u, v, G, L) the pollutant field is reconstructed as

$$x(p) = m(p)\ [\ \bar{x} + \Sigma_{k=1..r}\ \varphi_k\ a_k(p)\ ]$$

where $\bar{x}$ is the mean unit-norm shape over the training snapshots, φ_k is the k-th POD mode (a fixed spatial pattern), $a_k(p)$ is the modal coefficient giving how strongly mode k is activated for the input p, and m(p) is a scalar magnitude that rescales the unit-norm shape back into physical concentration units.

Coefficient model. At training time the "true" coefficient of each snapshot is its projection onto a mode, $a_k{}^{(c)} = \varphi_k{}^T V \hat{x}_c$. A light affine model is fitted once by least squares to reproduce these coefficients from the inputs, using the physics-informed feature vector f(p) = (u, v, G, L, |U|, 1/|U|, G/|U|) with $|U| = (u^2 + v^2)^{1/2}$, where L enters as the 0/1 corridor indicator - wind speed and emission-to-wind ratios enter because dilution scales with wind speed. Predicting a new case is then a single matrix–vector product.

Magnitude model and the RBF. The magnitude is written in multiplicative (log-additive) form so that emission changes follow passive-scalar physics:

$$\log m(p) = \log G + b_0 + b_L L + g(u, v), \quad L \in \{0, 1\}$$

The log G term encodes that, for a frozen flow field, concentration is exactly linear in the emission source strength. The b_L L term is a learned multiplicative corridor offset: it is switched on only when the intervention is active, and because the tagged segments are reduced by varying amounts (30–50%) its magnitude is fitted from the training snapshots rather than derived from a nominal reduction factor. The wind-dependent part g(u, v) is a thin-plate-spline radial basis function (RBF) interpolant, $g(u, v) = \Sigma_i \lambda_i \varphi(\|(u, v) - (u_i, v_i)\|)$ with $\varphi(r) = r^2 \ln r$, built over the training wind points because the same hourly winds recur across scenarios. The RBF captures the smooth but nonlinear way dilution depends on wind speed and direction, and interpolates naturally to unseen hours whose winds fall between the training points - far better than a single global slope in wind speed.

4.3 Validation

The number of POD modes is selected by comparing training and test reconstruction errors over a range of retained modes (Figure 1, left). The error decreases rapidly and then reach a plateau, indicating that the dominant spatial structures are already captured; five modes are retained as a compact and stable choice. Table 2 summarises the accuracy, and the parity plot (Figure 1, right) shows predicted vs reference cell values on the held-out set.

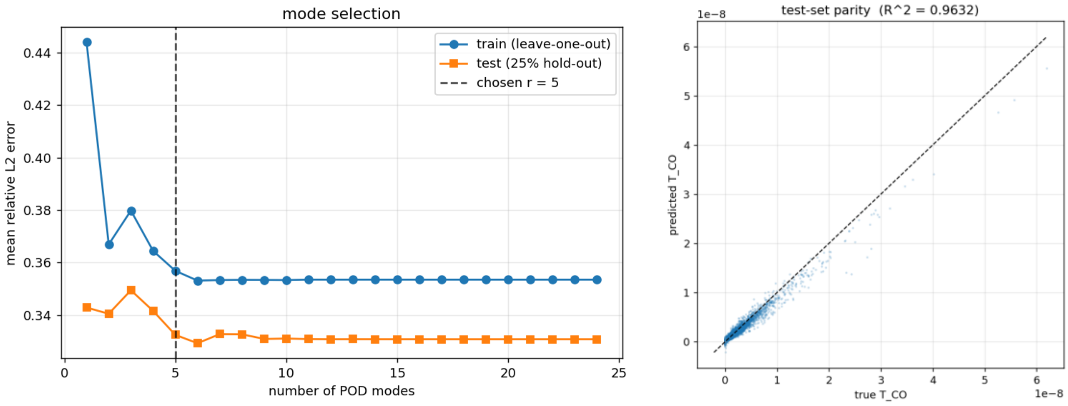

Figure 1. Left: mode-selection curve - reconstruction error vs number of retained POD modes (train leave-one-out and hold-out). Right: parity of predicted vs reference cell values on the test set (CO; NO_x is equivalent).

Pollutant	Modes r	Test relative-L2	Parity R^2
CO	5	0.332	0.963
NO_x	5	0.325	0.963

Table 2. Surrogate accuracy on the held-out test set.

Physical consistency check. In addition to the statistical split, the surrogate is checked against directly simulated data at receptor level. For the reference case at 07:00 the predicted CO field matches the CFD field to a volume-weighted relative-L_2 of 0.13, and the area-averaged concentration at the Public Hospital receptor is 1.52×10^{-9} vs the simulated 1.61×10^{-9} kg m^{-3} (~6%). This confirms that the reconstruction preserves the quantities used for the final assessment, not only the global field norm.

5. Results

5.1 Baseline exposure (CFD, ten sampled hours)

At baseline the most exposed receptor is Santos Simões (NO_x mean 6.5, peak 15.9 μg/m³), followed by Martins Sarmento and the Public Hospital; Francisco de Holanda is the least exposed. NO_x is the pollutant of interest, as CO is more than two orders of magnitude below its 8-hour guideline. Exposure peaks in the evening rush and at midday.

Receptor	CO mean	CO peak	NO_x mean	NO_x peak
Public Hospital	3.85	11.09	5.38	14.09
Francisco de Holanda HS	2.15	5.16	2.97	6.56
Martins Sarmento HS	3.76	9.70	5.52	12.60
Santos Simoes HS	4.46	11.23	6.47	15.91

Table 3. Reference concentrations (μg/m³), mean and peak over the ten sampled hours.

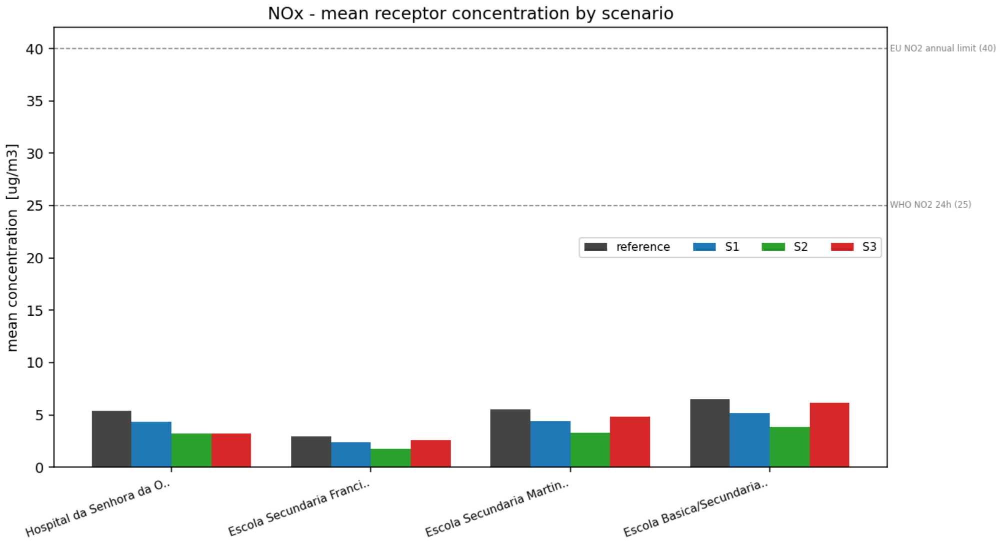

Figure 2. Mean NO_x at each receptor by scenario (dashed lines: EU/WHO NO_2 references; modelled NO_x is not ambient NO_2).

5.2 Scenario comparison (CFD)

Electrification scales uniformly - every receptor falls by exactly -20% (S1) and -40% (S2), for both pollutants and for both the mean and the peak. The Metro Bus is spatially selective: it removes only about 19% of the city-wide emission total yet cuts the Public Hospital by roughly-41%, while barely affecting the off-corridor receptors.

Receptor	Reference	S1	S2	S3
Public Hospital	3.85	3.08 (−20.0%)	2.31 (−40.0%)	2.26 (−41.2%)
Francisco de Holanda HS	2.15	1.72 (−20.0%)	1.29 (−40.0%)	1.89 (−12.2%)
Martins Sarmento HS	3.76	3.01 (−20.0%)	2.25 (−40.0%)	3.25 (−13.5%)
Santos Simoes HS	4.46	3.57 (−20.0%)	2.68 (−40.0%)	4.24 (−4.9%)

Table 4. CO - mean concentration (µg/m³) and change vs reference (ten sampled hours).

Receptor	Reference	S1	S2	S3
Public Hospital	5.38	4.30 (−20.0%)	3.23 (−40.0%)	3.21 (−40.4%)
Francisco de Holanda HS	2.97	2.38 (−20.0%)	1.78 (−40.0%)	2.62 (−11.9%)
Martins Sarmento HS	5.52	4.42 (−20.0%)	3.31 (−40.0%)	4.84 (−12.3%)
Santos Simoes HS	6.47	5.18 (−20.0%)	3.88 (−40.0%)	6.14 (−5.1%)

Table 5. NO_x - mean concentration (µg/m³) and change vs reference (ten sampled hours).

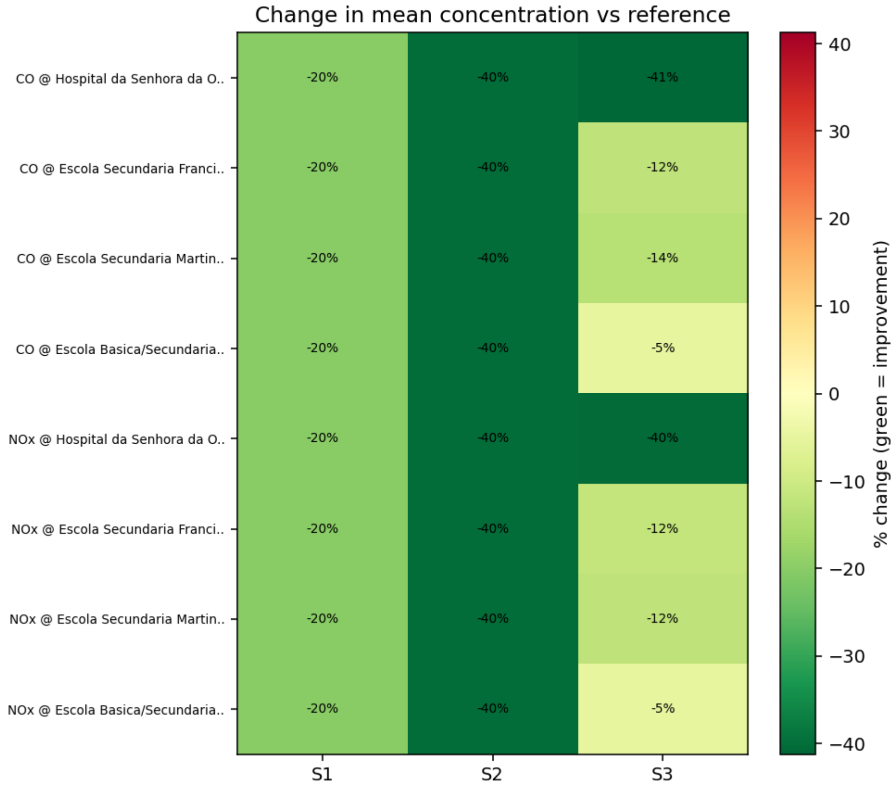

Figure 3. Change in mean concentration vs reference (green = improvement). S1/S2 uniform; S3 concentrated at the Hospital.

5.3 Spatial distribution

Placing the receptors on the Guimarães road network explains the scenario pattern. The S3 reduced corridor (blue) traces the Circular Urbana ring, which runs immediately next to the Public Hospital and far from Santos Simões. As such, so the Metro Bus removes the Hospital's dominant local source while leaving the most-exposed school essentially untouched. Road width is proportional to NO_x emission, marking the busiest source streets.

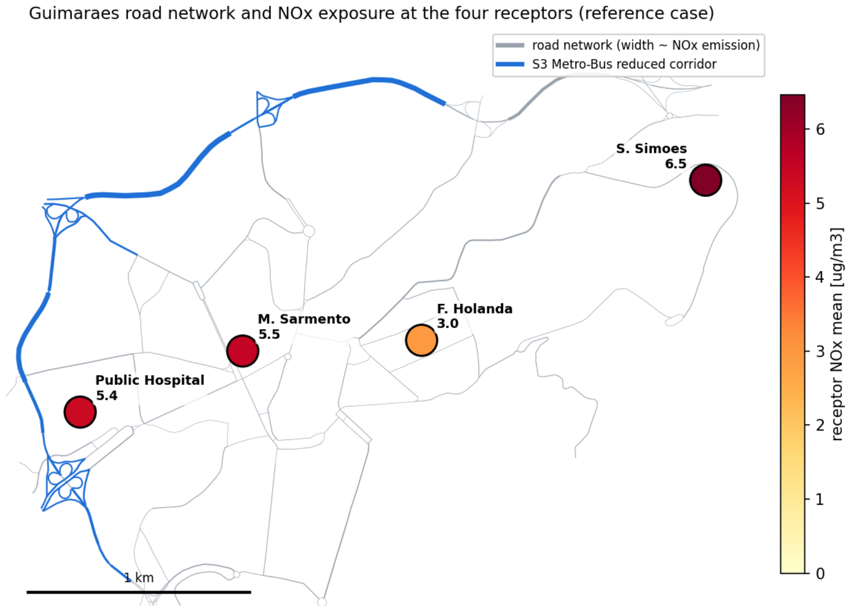

Figure 4. Road network (width ~ NOx emission), S3 reduced corridor (blue), and reference NOx at each receptor.

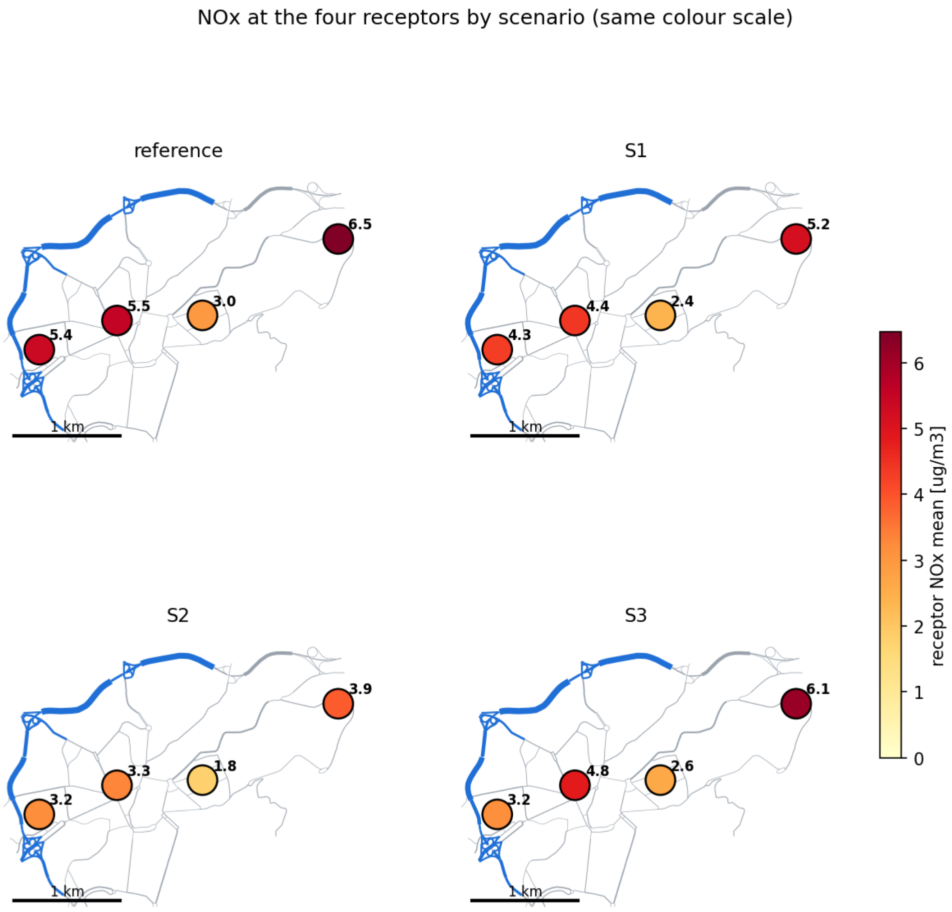

Figure 5. Receptor NOx by scenario on a shared colour scale: electrification lowers all sites uniformly; the Metro Bus (S3) acts mainly on the Hospital.

5.4 Full 24-hour prediction with the surrogate

The trained surrogate is evaluated over the complete daily sequence of wind conditions for all scenarios and both pollutants, and the reconstructed fields are sampled at the four sensitive receptors with the same post-processing definition as the original CFD workflow. This gives a full daily concentration profile at each receptor (Figures 6-7), with the daily-mean and peak values in Tables 6–7.

Receptor	Reference	S1	S2	S3	24h peak (Ref)
Public Hospital	3.68	2.94 (−20.1%)	2.20 (−40.2%)	2.58 (−30.0%)	9.54
Francisco de Holanda	2.04	1.63 (−20.0%)	1.22 (−40.0%)	1.71 (−15.9%)	5.28
Martins Sarmento	3.60	2.88 (−20.0%)	2.16 (−39.9%)	3.04 (−15.4%)	9.56
Santos Simões	4.34	3.47 (−19.9%)	2.61 (−39.9%)	3.86 (−11.0%)	11.66

Table 6. CO - daily-mean concentration (μg/m³) and change vs Reference over the full 24-hour cycle predicted by the surrogate.

Receptor	Reference	S1	S2	S3	24h peak (Ref)
Public Hospital	4.98	3.98 (−20.0%)	2.99 (−40.0%)	3.96 (−20.4%)	12.15
Francisco de Holanda	2.87	2.30 (−20.0%)	1.72 (−40.0%)	2.38 (−17.0%)	6.80
Martins Sarmento	5.35	4.28 (−20.0%)	3.21 (−40.0%)	4.45 (−16.9%)	13.00
Santos Simões	6.39	5.12 (−20.0%)	3.84 (−40.0%)	5.47 (−14.5%)	15.47

Table 7. NO_x - daily-mean concentration (μg/m³) and change vs Reference over the full 24-hour cycle predicted by the surrogate.

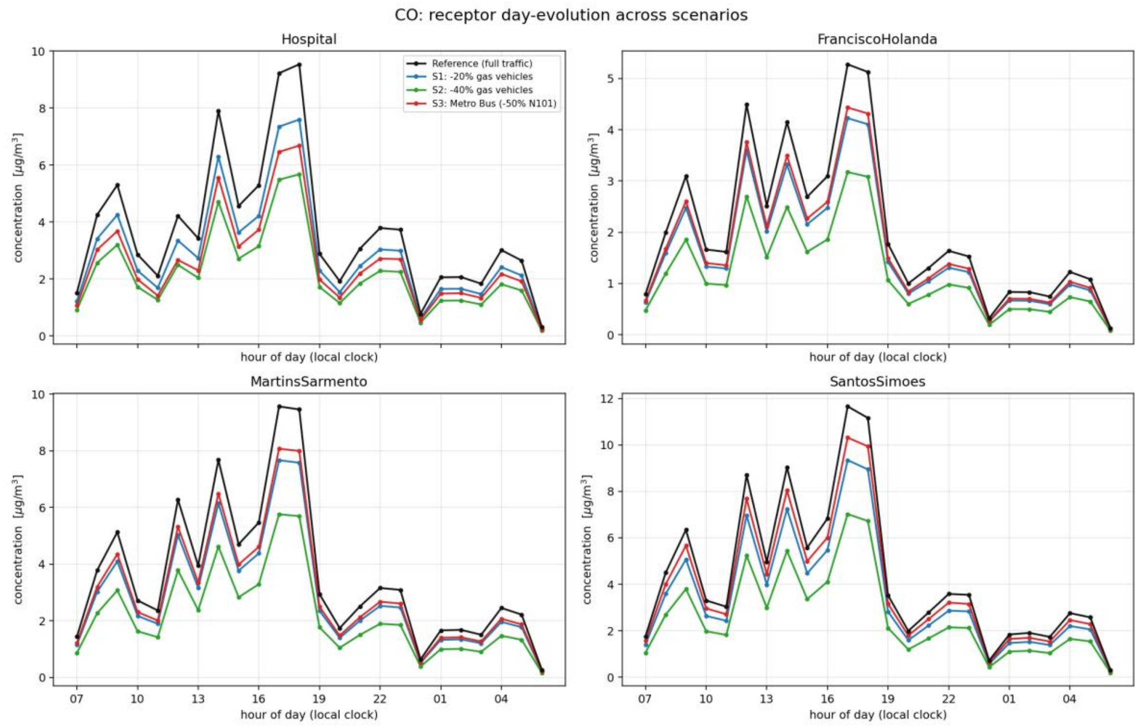

Figure 6. CO daily evolution at the four receptors (07:00 → 06:00), one curve per scenario. Reducing gas-based traffic (S1/S2) lowers every site uniformly, whereas the local corridor intervention (S3) is spatially selective.

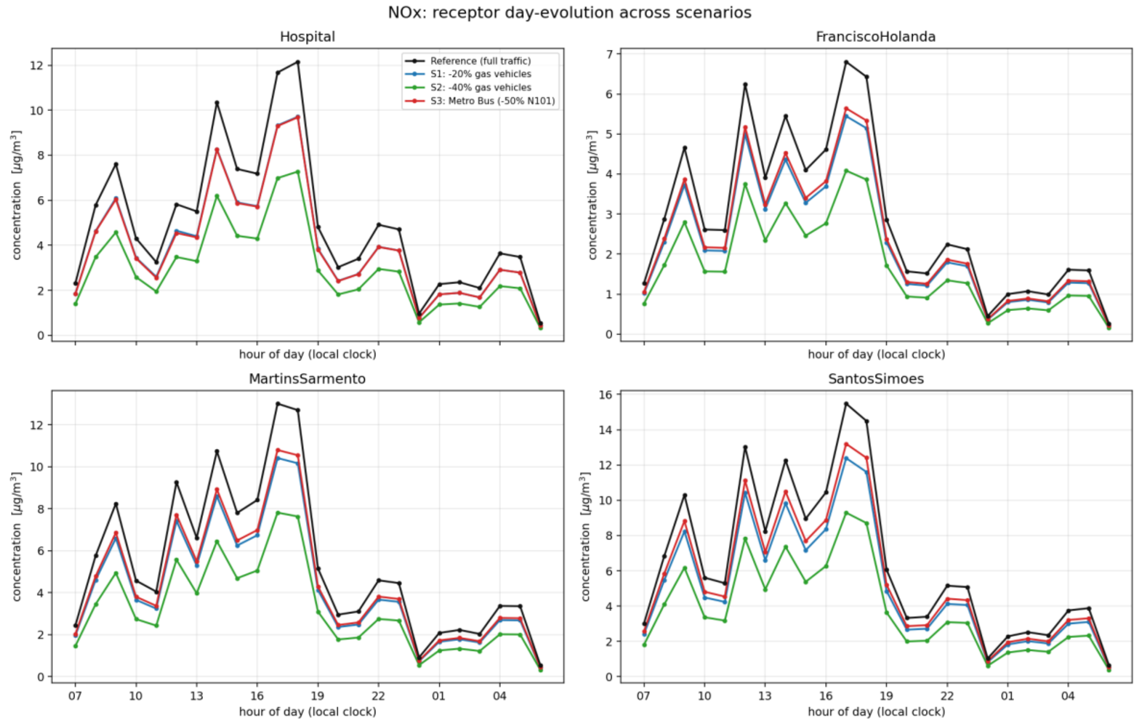

Figure 7. NOx daily evolution at the four receptors. The temporal trend follows the hourly wind cycle, with higher concentrations during less favourable dispersion periods.

Absolute levels and pollutant of concern. The predicted concentrations are the local traffic increment only (no regional background is added, and modelled NOx is not ambient NO_2), so they should be read as relative contributions rather than compliance values. On that basis the levels are modest: even the highest 24-hour peak (≈15.5 µg/m³ NOx at Santos Simões) sits below common short-term air-quality reference levels, and CO (at a few µg/m³) is negligible against its guideline, which is expressed in mg/m³. NOx is therefore the pollutant that governs the assessment, consistent with the direct CFD baseline of Section 5.1.

Temporal behaviour. The daily profiles show a strong diurnal swing driven by the combined traffic and wind cycle, with pronounced afternoon and evening-rush maxima and quiet overnight minima. Peak values are about 2–3× the daily mean at every receptor, so a daily mean substantially understates short-term exposure - which is precisely why the full 24-hour profile is more informative than a handful of sampled hours.

Scenario effectiveness over the full day. The surrogate confirms the CFD picture over the complete cycle: the global reductions (S1, S2) lower every receptor uniformly by −20% and−40%, because they act as a proportional cut of all sources, and even the stronger S2 case leaves Santos Simões the most exposed site (NOx mean still 3.84 µg/m³). The local corridor measure (S3) is the opposite: it is highly effective where it acts (30% CO and 20% NOx reductions at the Public Hospital, which borders the corridor) but delivers the smallest benefit at Santos Simões (-11% CO, -15% NOx). This is the key tension: the least costly measure does the least for the worst-exposed location.

Why S3 differs between the CFD and the surrogate. The two S3 estimates differ (about 40% NOx at the Public Hospital from the ten CFD hours, versus about -20% from the surrogate over the full 24 hours). This is a sampling-window effect, not an artefact of the corridor parameterization: the varying 30-50% segment reductions are fully represented in the snapshots the surrogate is trained on. The difference is specific to S3 because it is a local, spatially selective measure. The uniform electrification cases reduce every source by the same factor at every hour, so their -20/-40% change is identical in any hour and averages the same over any set of hours; the relative benefit of the corridor reduction, by contrast, varies from hour to hour, depending on how much of a receptor's concentration comes from the corridor under the prevailing wind and dispersion. The ten CFD hours were selected to represent exposure-relevant conditions and are weighed toward high-concentration, poorly ventilated hours, during which the neighbouring corridor dominates the Public Hospital and its removal gives a large reduction (about -40%). The surrogate instead averages over the complete daily cycle, including well-ventilated daytime and low-traffic overnight hours in which the corridor contributes a smaller share, so the day-averaged S3 benefit is diluted to about -20%. Both figures are correct for their window: the CFD value reflects the high-exposure hours that matter most for health, while the surrogate value is the true daily-mean reduction.

5.5 Sensitivity to the sampling method

The receptor metric above is the average area over each ROI surface. Recomputing the same runs with a volume average over a box of breathing air above each receptor gives the table below. Under both metrics the electrification scenarios fall by exactly -20% (S1) and -40% (S2), a linear response, and the spatially selective S3 agrees to within about 2 percentage points at every receptor, so the scenario conclusions do not depend on the sampling choice. The absolute levels differ modestly, and the two metrics even rank the most-exposed site differently: the surface average puts Santos Simoes highest, whereas the volume average puts the Public Hospital highest.

Receptor	ref (surface)	ref (volume)	S3 (surface)	S3 (volume)	ref (surface)
Public Hospital	5.38	6.4	−40.4%	−42.1%	5.38
Francisco de Holanda	2.97	2.6	−11.9%	−13.5%	2.97
Martins Sarmento	5.52	6.2	−12.3%	−14.4%	5.52
Santos Simões	6.47	5.2	−5.1%	−4.2%	6.47

Table 8. NOx mean (μg/m³) and S3 change vs reference under the two sampling metrics; S1/S2 are −20/−40% under both.

5.6 Influence of the computational domain

The receptor results were also compared between two computational domains: a small COST-732 domain (about 6.3 km footprint, cylinder radius 3000 m, ~1.7 M cells) and a large city4CFD domain (about 25 km, ~40 M cells, with a porous vegetation canopy). For the reference case at a matched hour (data hour 1), the surface area-average receptor concentrations are:

Receptor	NOx small	NOx large	CO small	CO large	NOx small
Public Hospital	15.45	6.16	10.38	4.34	15.45
Francisco de Holanda	15.89	2.76	11.49	2.04	15.89
Martins Sarmento	23.01	6.00	15.80	4.16	23.01
Santos Simões	10.32	6.33	7.29	4.38	10.32

Table 9. Domain influence: the small domain over-predicts receptor concentrations and reorders their ranking (indicative, see caveat).

Finding. The small domain over-predicts every receptor, by roughly 1.6-5.8× for NOx (largest at Francisco de Holanda), and it reorders the ranking: the small domain makes Martins Sarmento by far the most exposed and Santos Simoes the least, whereas the large domain flattens the four receptors to around 6 μg/m³. The tight 3000 m domain has a higher blockage ratio and boundaries closer to the source region, which restricts dilution and traps pollutant, whereas the 25 km domain gives the approach flow far more fetch to disperse. Reassuringly, the scenario percentage-changes are far less domain-sensitive, since they are set by the linear emission scaling rather than the domain, so the scenario conclusions carry across domains even though absolute levels do not.

Caveat. This single-hour comparison also differs in the vegetation treatment (the large domain includes the porous canopy, the small one does not) and in mesh resolution, so it mixes domain extent with those effects and should be read as indicative; a fully controlled study repeats the small domain over all ten hours with identical settings.

5.7 Ground-level concentration field

Figure 8 is a top-down view of the ROI with the terrain and street surfaces coloured by near-ground NOx (reference case, log scale); the buildings are shown in black and the four receptor areas (ROIs) in green. The plume is concentrated along the road network (strongest on the Circular Urbana ring and through the dense city core) and falls off toward the urban periphery, consistent with the receptor pattern of Sections 5.1-5.3.

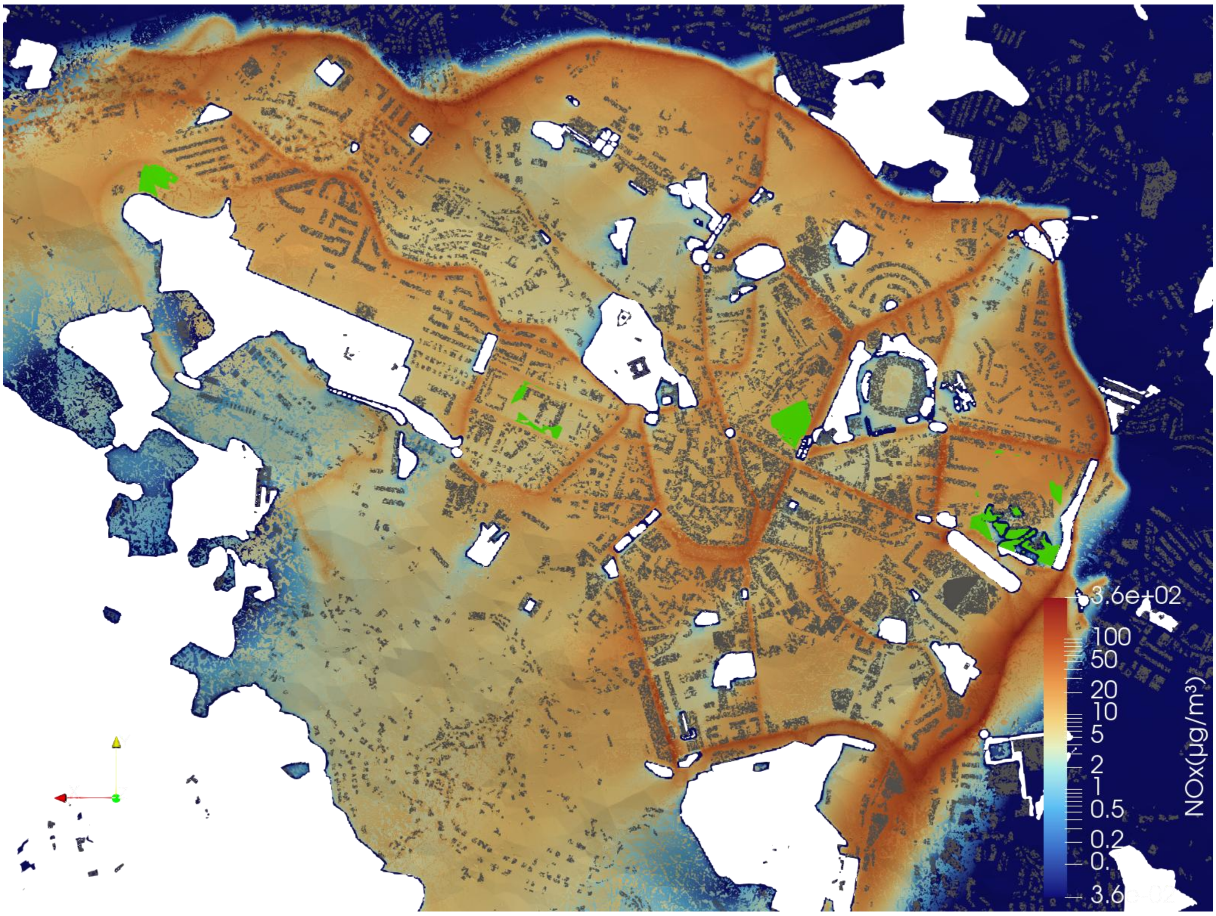

Figure 8. Top-down view of near-ground NOx over the terrain and streets (reference case, μg/m³, log scale). Buildings are shown in black and the four ROI receptor areas in green; warm colours trace the traffic plume along the street network.

6. Discussion and conclusions

The three measures serve different objectives. S2 (40% electrification) is the strongest broad-based measure, lowering every sensitive receptor uniformly by 40%; S1 gives the same effect at half the magnitude. The Metro Bus is the most cost-effective intervention for the Public Hospital specifically, since it achieves near-S2 benefit there while changing far less traffic overall, because the reduced corridor is the Hospital's dominant local source, but it does little for receptors away from the corridor and almost nothing for Santos Simões, which is the most exposed site at baseline. This spatial pattern, first observed in the ten directly simulated hours, is confirmed by the surrogate over the full 24-hour cycle.

The practical conclusion is that no single measure is best everywhere. A combined strategy- the Metro Bus to protect the Hospital and the central corridor, plus partial fleet electrification to bring down the off-corridor schools, above all Santos Simões. This would deliver the most uniform city-wide improvement. If a single receptor must be prioritised, the choice of measure depends on which one: the bus for the Hospital, electrification for Santos Simões.

Two consistency results give confidence in the modelling chain. First, the exact -20%/-40% response of the electrification scenarios at every receptor confirms that dispersion is linear in emission for a frozen flow, so the emission mapping and the reuse of one flow field per hour work as intended. Second, the PODI surrogate reproduces held-out CFD fields with a parity R^2 of 0.96 and matches receptor-level CFD values to within ~6%, demonstrating that the existing CFD dataset can be transformed into a fast and reliable reduced-order model. Because each surrogate evaluation is essentially free, the full daily exposure cycle, continuous sweeps of the reduction level, and combined scenarios can all be screened without repeating the CFD workflow.

7. Limitations and future work

The direct-CFD statistics are over ten representative hours (the same hours for every scenario), while the 24-hour figures come from the surrogate; concentrations exclude regional background and modelled NO_x is not ambient NO_2, so absolute levels should be read as relative traffic increments rather than compliance values. The surrogate inherits the assumptions of the underlying CFD (steady RANS, frozen flow per hour, passive-scalar dispersion) and its accuracy is bounded by the snapshot coverage of the wind and emission space.

Natural extensions are: a combined Metro-Bus-plus-EV scenario and a continuous sweep of the reduction level, both essentially free with the surrogate; daily aggregated exposure metrics over the true 24-hour cycle; full spatial concentration maps of the corridor to complement the receptor numbers.

8. Code and data availability

All OpenFOAM cases, tools, post-processing scripts and workflows developed for this challenge are openly available in the project repository. The repository documents the meshing, flow and dispersion cases, the emission-mapping and receptor-sampling tools, the Snakemake POD-snapshot workflow, and the reporting pipeline, so the jury can inspect the workflow and reproduce the main results.